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Conceptualizing Cloneable Automation

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ABSTRACT

This thesis conceptualizes cloneable automation and examines which attributes and prerequisites can make an automation solution possible to replicate across contexts with limited re engineering and remaining usable for its intended users. The study is based on an exploratory qualitative design guided by Design Research Methodology. The work combines a literature study with semi structured interviews with automation experts, two site visit at the case company, and recurring case related discussions. The final case oriented part of the thesis is treated as a prescriptive study, where the descriptive understanding is translated into tentative guidelines for PLC, robotics, and HMI.

The result shows that reuse in current automation practice already exists, but most often as partial reuse of base programs, libraries, HMI structures, and earlier concepts rather than as full instantiation of a finished solution. The most time consuming work is concentrated in PLC and software logic, robot application logic, communication and interface integration, HMI adaptation, commissioning, safety work, validation, and the installed base. The second site visit material supports this result and also show that practical cloneability depends on company ownership of code and knowledge, mandatory documentation as part of the delivery workflow, reusable PLC and robot blocks, a stable core with bounded local variation through parameters and recipes, and standards that remain teachable and usable.

In this thesis, attributes define what cloneability is, while guidelines explain how those attributes can be operationalized in PLC, robot, and HMI design. The main conclusion is that cloneable automation is described as engineered reuse built around a stable, documented, bounded, and usable core rather than copy based reuse of duplicated artefacts.

(Keywords: Cloneable automation, automation reuse, modular automation, reconfigurable manufacturing, PLC, robotics, HMI, parameterization)

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ABBREVIATIONS

The following abbreviations are used in the thesis.

AML	Automation Markup Language
CE	Conformité Européenne
DRM	Design Research Methodology
FAT	Factory Acceptance Test
HMI	Human-Machine Interface
IIoT	Industrial Internet of Things
ML	Machine Learning
MTP	Module Type Package
OEE	Overall Equipment Effectiveness
OPC UA	Open Platform Communications Unified Architecture
PEA	Process Equipment Assembly
PLC	Programmable Logic Controller
PnP	Plug and Produce
POL	Process Orchestration Layer
RMS	Reconfigurable Manufacturing System
SAT	Site Acceptance Test

1. INTRODUCTION

This chapter introduces the background and problem area of the thesis. It explains why manufacturing companies need automation solutions that can be adapted and reused with less effort, and why cloneable automation becomes relevant in this context. The chapter also presents the problem formulation, aim, research questions, and the limitations that frame the work.

1.1. Background

Manufacturing companies increasingly face requirements for rapid change, shorter product cycles, and higher product variety (Wallstersson & Öberg, 2025). To meet this demand, research and industry have tried to invent production and automation concepts that support reconfigurability, reuse, and fast adaptation rather than rigid, one off solutions. One established perspective is reconfigurable manufacturing, where systems are designed to repeatedly change capacity and functionality in a cost efficient way, enabled by characteristics such as modularity and integrability (Andersen et al., 2017).

The challenge is not only to make production systems technically flexible, change often means additional engineering, integration, commissioning, and validation work. Setup costs are closely tied to the time required for planning, software and hardware configuration, and program adaptation, while competence needs may also require both in-house and external expertise (Ramasamy et al., 2023). When change work becomes slow or competence intensive, development time increases, modifications are delayed, and the business case for automation becomes harder to motivate, especially when expected gains are uncertain or when evaluation and decision-support activities themselves are difficult and time-consuming to carry out (Andersen et al., 2017).

Against this background, there is a growing need for automation solutions that can be deployed and adapted with less effort and by a broader set of users and not only specialists. For example, recent work on, smart portable automation, frames the demand for flexible and reconfigurable automation that can be utilized by non experts as an important direction for maintaining competitiveness (Wallstersson & Öberg, 2025).

This thesis builds on the idea that reusability should not only apply to mechanical modules, but also to the practical engineering work of automation, software, interfaces, configuration, commissioning, and day to day adjustments. The MARC project behind this work has created the idea of cloneable automation, where an automation solution should be possible to replicate across contexts with minimal re engineering. Without such an approach, companies risk remaining dependent on one off engineering solutions that are costly to scale, difficult to transfer, and hard to use in organizations with limited specialist competence. In this sense, cloneability is not only a technical ambition, but also a practical condition for making reusable automation workable in industry.

1.2. Problem formulation

Cloneable automation lacks a clear definition and a shared set of attributes, without such a definition, it becomes difficult to specify requirements, compare approaches, and evaluate

whether an implemented solution is cloneable in practice, which limits how easily the solution can be reused or relocated in different contexts.

Also, technical advances can still fail to deliver value when systems are too complex or when users do not know how to use available functionality.
This highlights a key socio technical challenge, even if a system is technically portable, it may not be practically cloneable if it requires expert interventions for small adjustments, or if it increases perceived workload for operators and technicians.

The implement ability of cloneable automation may differ across organizational contexts. Large companies may have access to specialized competence somewhere in the organization, while smaller companies may not, meaning that the same solution can be easy in one context and unrealistic in another.
Together, these issues motivate a need to define cloneable automation in a way that is both technically meaningful and practically usable.

1.3. Aim and Research questions

The aim of this thesis is to conceptualize cloneable automation and identify the attributes and prerequisites that enable an automation solution to be replicated across contexts with minimal re-engineering while remaining usable for its intended users.

To reach this aim, the thesis addresses the following research questions:

RQ1. What characterizes a cloneable automation system, and how does it differ from traditional replication or duplication of manufacturing resources?

RQ2. What building blocks in an automation system consume the most engineering and/or integration time?

RQ3. Which attributes define and enable clonability?

RQ4. How can the identified clonability attributes be broken down into design guidelines for the most important building blocks of an automation system?

1.4. Project limitations

This thesis is conducted within the scope of a 30-credit master's project and is therefore limited in time, scope, and empirical coverage. The study focuses on conceptualizing cloneable automation systems by defining and clustering attributes that enable reuse and replication, and by applying these attributes to one under development assembly line concept for product handling in a single case company, focus will be on the assembly processes. One interesting enabler for cloneable automation can be machine learning, it is mentioned but will be left as an option for future research. Safety and validation is another important area that is mentioned in the thesis, also this will be left for future research.

2. RESEARCH METHOD

This chapter describes how the study was carried out. The research is based on an exploratory qualitative design, where a literature study, expert interviews, one site visit, and recurring case related discussions are combined. The chapter explains how the material was collected, how the cloneability attributes were identified, and how quality and ethical aspects were considered.

2.1. Research design

This thesis has an exploratory qualitative design with a single case orientation. The aim is to conceptualize cloneable automation and identify the attributes and prerequisites that make repeated reuse possible in practice. The study combines a literature study with qualitative empirical material from interviews, meetings, contextual observations, and case related discussions. The aim of this thesis is not to test a mature theory, but to conceptualize cloneable automation as an emerging concept and examine how it can be translated into one industrial context.

The research approach is guided by Design Research Methodology (DRM) by Blessing and Chakrabarti (2009). Within that framework, the thesis most closely resembles a Type 2 project, where a review-based Research Clarification is followed by a comprehensive Descriptive Study I and an Initial Prescriptive Study. In this thesis, the Research Clarification stage was used to define the problem, aim, research questions, and delimitations. The Descriptive Study I stage then combined literature and empirical material to build the concept of cloneable automation and identify its attributes.

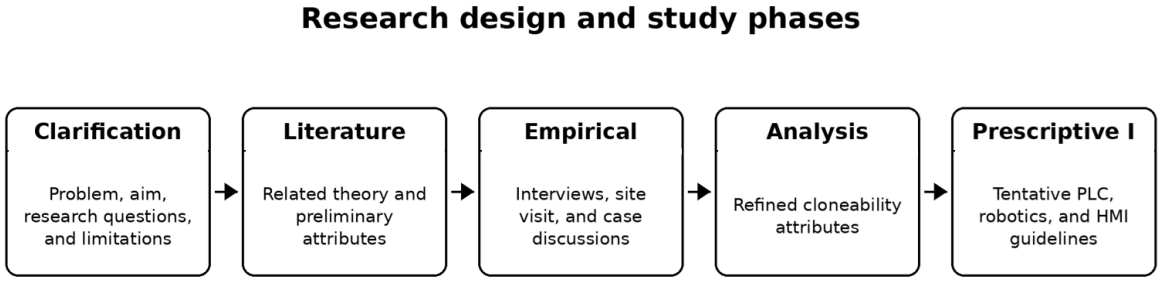


Figure 1: Overview of the research design and study phases. Figure created by the author.

Figure 1 gives an overview of how the phases of the thesis relate to the DRM logic. The literature study and the empirical material support the descriptive understanding of the problem, while the final case oriented interpretation works as an Initial Prescriptive Study where the cloneability attributes are used to form tentative design guidelines for PLC, robotics, and HMI.

The final case oriented part of the thesis is treated as a prescriptive study. In this part, the descriptive understanding from literature, interviews, site visits, and case discussions is translated into tentative design guidelines for the automation areas where cloneability has to be operationalized in practice. The prescriptive contribution is not a finished universal method, but a grounded proposal for how the identified attributes can guide PLC, robot, and HMI design in the studied case context.

The study is directed towards one under development assembly line concept for product handling, with focus on assembly processes. The work began with introductory meetings with the industrial supervisor and researchers connected to the project. These meetings gave an initial understanding of how cloneable automation was discussed in practice and what technical and organizational challenges were already visible. In the start of the thesis, one site visit was carried out at the case company in the start of the thesis, where the underdeveloped assembly station and its surrounding production context were presented. Weekly supervision meetings were then used throughout the thesis process to discuss progress, challenges and plan the next steps.

A second site visit was carried out near the end of the thesis to strengthen the case oriented part of the study. This visit focused more directly on how cloneability can be made practical through company owned code, documentation routines, reusable PLC and robot blocks, robot standards, stable core structures, parameterized variation, and teachable standards across different competence levels. The material from this visit was used mainly to refine the results and to make the later design guidelines more operational.

2.2. Literature study

A literature study was carried out to identify concepts, proposed attributes, enabling mechanisms, standards, and trade-offs relevant to cloneable automation. Since cloneable automation is not an established research term, the search was broadened to adjacent areas such as modular automation, PnP, reconfigurable manufacturing, brownfield integration, and human factors. This step was necessary since the concept had to be built through related literature rather than extracted from one existing research stream.

Searches were done in ScienceDirect, Scopus, IEEE Xplore, Sage Journals, MIS Quarterly, ResearchGate, DiVA, the MDU Library Primo search engine, and Google Scholar. The search terms were grouped around four main themes:

1. cloneable, replicable, reusable, and scalable automation

2. modular automation, process equipment assemblies, modular production, and plug and produce

3. interfaces and standards such as OPC UA, Module Type Package, Process Orchestration Layer, and AutomationML

4. legacy systems, brownfield integration, human factors, perceived complexity, and organizational maturity.

Backward snowballing from the reference lists of key papers was used to identify further relevant sources. To ensure reliability, peer reviewed journal articles and conference papers were selected.

Only papers with downloadable pdf were selected. The first screening was done from the title, followed by the abstract, and then by full text reading when a publication appeared relevant to the aim of the thesis. The collected literature was organized in EndNote and grouped by topic. This made it easier to compare arguments across sources and return to the original publications during later analysis and writing.

The DRM book by Blessing and Chakrabarti (2009) was used specifically to structure the research approach and to clarify how the thesis relates to different stages and research types in design research.

2.3. Empirical data collection

The empirical data was collected using semi structured interviews with automation experts, two site visit at the case company, and recurring supervision and project discussions to guide the thesis. The idea with the empirical data collection was to challenge and refine the attributes identified in the literature. Another idea, was to understand what makes cloneability difficult or valuable in real applications, especially in relation to engineering effort, programming effort, integration, reuse, and adaptation.

The interviewees were selected based on expert knowledge from the field, for example, hands-on experience from automation engineering, system integration, robotics, and IIoT related work. The interviews were important because the thesis deals with an emerging concept where expert knowledge about implementation work is highly relevant. The interviews were semi-structured in order to have a more open and natural conversation, probe questions were used to dig deeper when needed to expand where interesting ideas where expressed. Most interviews were recorded and automatically transcribed in Microsoft Teams, which made it possible to keep the interviews more natural.

Name	Profession	Duration	Type of interview
Automation Expert 1	Automation engineer	31 min	Semi Structured
Automation Expert 2	System Integrator and IIoT expert	35 min	Semi Structured
Automation Expert 3	Automation Engineer	38 min	Semi Structured
Automation Expert 4	Automation Engineer	60 min	Semi Structured
Automation Expert 5	System Integrator, Automation Engineer	28 min	Semi Structured
Automation Expert 6	System Integrator	55 min	Semi Structured
Automation Expert 7	System Integrator	44 min	Semi Structured

Table1: Interviewees

The interviews lasted between 28 and 60 minutes. The interview guide was around how reuse is currently achieved in automation projects, what causes repeated engineering effort, which conditions make an automation solution easier or harder to replicate, and how such conditions may differ across organizational contexts.
In addition to the interviews, one early site visit was carried out at the case company, where the under development assembly line concept and its surrounding production context were presented. This visit was used to understand the technical scope of the case and the context in which cloneability would later be discussed. Weekly supervision meetings were then used throughout the thesis to discuss the relevance of attributes, and plan the next steps.

The second site visit added a more implementation oriented empirical material. The collected material consisted of visit notes and reflection notes. The material focus on code ownership, commenting and documentation, versioning, cybersecurity as part of the automation standard, reusable PLC function blocks, reusable robot structures, naming conventions, stable core logic, local variation through parameters and recipes, and the need to make standards understandable for different user groups.

2.4. Identification of cloneability attributes

Since cloneable automation is not an established concept with a shared definition, the attributes used in this thesis were identified through a literature review of adjacent research areas relevant to repeated reuse of automation solutions, for example, modular automation, PnP, reconfigurable manufacturing, interoperability, brownfield integration, and human factors. The selected literature was grouped into themes, and recurring concepts describing conditions that reduce re engineering effort while preserving practical usability were identified. In the next step, these recurring concepts were selected as preliminary attribute candidates.
The preliminary attributes were discussed on supervisions and was suggested to be linked to the main areas, PLC, HMI and robotics.

In the analysis, a distinction was made between cloneability attributes and design guidelines. The attributes describe what defines or enables cloneability at a conceptual level, for example a stable reusable core, modular boundaries, standardized interfaces, bounded parameterization, documentation, and usability. The design guidelines describe how these attributes can be operationalized in the main automation areas, especially PLC, robot, and HMI.

2.5. Analysis of empirical material

The empirical material was analyzed thematically with the aim of identifying recurrent patterns in how the interviewees described reuse, engineering effort, integration, commissioning, legacy constraints, validation, usability, and cloneable automation. The interviews formed the main basis for the analysis, while observations from the two site visit and recurring supervision and project discussions were used as contextual support in the interpretation. The analysis did not follow a formal line by line coding procedure, instead the material was read repeatedly and organized around broader recurring themes that were relevant to the research questions.

The themes were partly guided by the structure of the semi structured interview guide, which functioned as an initial thematic frame, and partly refined through recurrent issues that appeared across several interviews. This means that the analysis had a mainly deductive structure, since it followed the central topic areas of the interview guide, but also an inductive element because the final formulation of the themes was based on patterns that recurred in the empirical material. The purpose was not to quantify how often something was mentioned, but to identify similarities, differences, and recurring conditions that could contribute to the conceptualization of cloneable automation and the identification of cloneability attributes.

2.6. Quality considerations and research ethics

To ensure quality of the study, the thesis relies on triangulation by combining literature, expert interviews, contextual observations from the two site visit, and repeated analytical discussions around the case. This reduced the risk of building the argument on a single source or a single

viewpoint. Also, interview summaries were sent back to participants for checking when needed, which reduced the risk of misunderstanding.
Also, interviews where transcribed, notes where taken during supervision discussions, this made it possible to return to the original material during analysis. This thesis focuses on a, single case orientation which limits statistical generalization, but that is not the aim of the thesis. The aim is more towards analytical generalization, to develop a clearer conceptualization of cloneable automation and show how the concept can be interpreted in one industrial context.
Regarding ethical concerns, the interviewees are presented anonymously in the thesis, and the case related part of the analysis is written at a level where the focus remains on methodological and conceptual insights rather than on unnecessary company specific detail.

3. THEORETICAL FRAMEWORK

This chapter presents the theoretical framework used to understand cloneable automation. Since cloneable automation is not an established academic concept, related areas such as Plug and Produce, modularity, reconfigurable manufacturing, interoperability, legacy systems, and human usability are used to build the conceptual background. The chapter also develops a working definition and a first concept model for cloneability attributes.

3.1. Why cloneable automation

Manufacturing systems are increasingly expected to handle high product variety and rapid change, which reduces the time available for engineering and ramp up. Shorter product life cycles and individualized customer requirements intensify this pressure. Plug and Produce (PnP) is positioned as an approach to reduce engineering time when production lines must be changed rapidly (Um et al., 2017). In the same direction, individualized production challenges established manufacturing approaches in terms of modularization, flexibility, and efficient reconfiguration, while the business driver is framed as minimizing the time from objective definition to start of production (Wrede et al., 2016). Together, these drivers push industry toward production systems that can be adapted and scaled repeatedly rather than optimized once for long periods.

When it comes to flexibility, it is not only a property of the physical line, it is also shaped by the engineering effort required to implement change. In highly automated systems, modifications often demand substantial integration and configuration work, and the effort becomes visible when new products are added or existing ones are modified. The shift from mass production to mass customization increases product variety, and long change times translate into losses through production downtime (Michalos et al., 2016). An example where long change times are present is in robotized stations, even if robots are inherently flexible, the installation, setup, and integration of a robot with station level control can take significantly longer than for other machines because multiple aspects must be configured, including PLC integration (Michalos et al., 2016). Here the limiting factor is often not the robot hardware itself, but the integration and commissioning effort within existing control and infrastructure constraints.

PnP concepts comes with integration challenges, production systems are highly interdependent, meaning changes in one part can propagate to other subsystems and processes, and the configurations needed to integrate new technologies, machines, and robots are still largely developed manually, often time consuming, error prone, and competence intensive (Fattouh et al., 2023). Similar patterns are reported in complex cyber physical monitoring settings, where key setup and design work remains manual because developers must select, parameterize, operate, and validate sensors and evaluation algorithms across many possible solution variants, while expert judgement is still needed to understand the monitored system and interpret monitoring decisions (Mönks et al., 2016). When integration remains specific and largely manual, scaling solutions, such as replicating a workcell or extending a line, tends to reproduce similar engineering workload and integration risks.

This highlights a core trade-off in the thesis, production systems need more flexibility to handle change, but adding flexibility often makes the system more complex and increases the engineering effort. Wrede et al. (2016) describe this as the challenge of increasing flexibility while still controlling the inherent complexity that comes with reconfigurable systems and tight

vertical integration. PnP is similarly described as aiming to automate integration work while also introducing new system complexity and a continued need for competencies and configuration validation to ensure correctness (Fattouh et al., 2023). Against this background, the relevance of cloneable automation is that it reframes flexibility as a repeatability problem in automation engineering, the goal is not only to reconfigure once, but to repeatedly deploy and adapt automation solutions across contexts with predictable effort and limited re-engineering. Within modular production ecosystems, higher reusability of readily available modules is argued to enable new processes to be implemented more easily, faster, and at lower cost, with indications that only a smaller subset of components may need to be engineered from scratch as the ecosystem matures, faster commissioning in new scenarios is linked to reusing hardware, software components, and sub processes (Wrede et al., 2016).

3.2 Concepts related to cloneable automation systems

This section presents concepts that are used to understand cloneable automation. Since cloneable automation is not an established concept, related concepts such as Plug and Produce, modularity, reconfigurable manufacturing, bounded flexibility, and human usability are used to explain how automation solutions can be reused across contexts with limited re engineering.

3.2.1 Plug and Produce

PnP aims to integrate production resources so adding or replacing equipment requires minimal engineering effort and short downtime. PnP extends the plug and play ideas from the computer world to manufacturing and is now linked to modular, reconfigurable production (Arai et al., 2000). One attribute that makes PnP powerful is that it relies on self-configuration, one idea is that new components should be integrated through configuration and parameterization rather than ad hoc coding (Bennulf et al., 2022; Khabbazi et al., 2023).

A common realization is distributed control, often multi agent or holonic, where resources and parts are represented as interacting entities, this supports flexible coordination and has been demonstrated in shop floor reconfiguration with module addition or removal without programming effort (Khabbazi et al., 2023; Onori et al., 2012).

Self-configuration is strengthened by machine readable capability and interface models. Ontology based methods can reason over capabilities and interface relations to derive practical configurations, while skill modeling and service oriented orchestration describe resource skills and support dynamic composition of services (Eymüller et al., 2023; Pfrommer et al., 2015).

3.2.2 Modularity and reconfigurable manufacturing

Reconfigurable manufacturing systems (RMS) are commonly presented as a response to higher product variety, shorter product life cycles, and frequent changes in demand. In this view, reconfigurability is the ability to change system structure and reconfigure hardware and software to adjust both capacity and functionality (Bortolini et al., 2018). Modularity is described as breaking production functions into separate modules that can be rearranged. Integrability means that these modules can be connected quickly and reliably through compatible mechanical, data, and control interfaces (Bortolini et al., 2018).

One enabling mechanism is to decouple goals from specific resources. In PnP, goal oriented multi agent approach, process plans can be expressed through required skills and interface types, and the mapping to a concrete resource is resolved at runtime based on the current

system situation (Nilsson et al., 2023). Physical reconfiguration is supported when modules and resources rely on standardized interface contracts, for example standardized mechanical, fluidic, and signal connections or standardized energy and media interfaces on process modules (Markaj et al., 2022; Nilsson et al., 2023). In modular automation, encapsulation is achieved by letting each module provide its process functions as services, described in a vendor-independent MTP that the Process Orchestration Layer can import and use to control the module without depending on the module's internal controller implementation (Hoernicke et al., 2020). One aspect to highlight here, even with the same MTP concept, vendors can design services differently, for example, different scope, procedures, parameter sets, state and behavior assumptions, which can reduce true interoperability because the meaning and behavior of the same service may not match across implementations (Bittorf et al., 2023).

The practical benefits of modularity come with trade-offs that matter for engineering and operations. RMS literature links dynamic markets and rapid technology introduction to higher structural and dynamic complexity, and hybrid modular monolithic plants further increase the solution space and integration complexity (Bortolini et al., 2018; Markaj et al., 2022). A second trade-off concerns performance: modular plants may offer higher flexibility but can lack optimality compared with tailored monolithic equipment (Markaj et al., 2022). At the control level, decentralized approaches may suffer from decision myopia and limited operational performance guarantees for real-time control (Ribeiro et al., 2015; Scrimieri et al., 2022). Reconfiguration can also create overhead and compliance constraints, for example information-processing load, the need to regulate system stability, and safety assessments after each reconfiguration (Koo et al., 2018; Ribeiro et al., 2015).

For this thesis, modularity and reconfigurability provide a concrete reference point for what repeatable change requires in practice. The reviewed work points to interface standardization, service based encapsulation, for example, MTP, POL and skill based decoupling with runtime binding as key mechanisms for reducing re-engineering effort (Bortolini et al., 2018; Hoernicke et al., 2020; Nilsson et al., 2023). At the same time, practical cloneability must manage complexity and performance trade-offs and treat safety and legacy integration effort as part of the engineering cost of (Koo et al., 2018; Markaj et al., 2022; Ribeiro et al., 2015).

3.2.2.1 Flexibility vs reconfigurability

In this thesis, flexibility and reconfigurability are treated as related but distinct concepts. Flexible manufacturing systems (FMS) are typically described as having built in flexibility to handle expected product variants, meaning the system is designed upfront to manage predefined variation (Koo et al., 2018; Scrimieri et al., 2023). This aligns with the view that flexibility can be embedded in a traditional control solution through preprogrammed logic, but that such flexibility primarily covers foreseen needs (Nilsson et al., 2023).
Reconfigurability, on the other hand, emphasizes the capability to change the system itself. In the RMS literature, reconfigurability is described as the ability to rapidly change structure and hardware, software configuration in order to adjust production capacity and functionality (Bortolini et al., 2018). Scrimieri et al. (2023) further distinguish RMS from FMS by highlighting scalability enabled through modular equipment, rather than relying on extensive built in functionality.

Here follows explanation and examples to separate the terms:
Flexibility: Handling variation within an existing configuration, predefined product variation.

Reconfigurability: Changing the configuration or structure to achieve new capacity and/or functionality.

For example, focus on product variation: A system is flexible if it can produce predefined variants by switching programs, recipes within the same setup (Koo et al., 2018; Nilsson et al., 2023). A system is reconfigurable if it can be changed structurally, for example, by rearranging or adding modular equipment, to extend functionality or scale capacity when demand or requirements shift (Bortolini et al., 2018; Scrimieri et al., 2023).

Another example, with focus on cost and design intent: FMS can become costly when flexibility is achieved through "excessive built-in functionalities" aimed at predefined variation (Koo et al., 2018). RMS instead targets adaptability and scalability by reconfiguring modular resources as needed (Bortolini et al., 2018; Scrimieri et al., 2023).

3.2.2.2 Degree of flexibility, balancing flexibility with complexity and engineering effort

Flexibility is repeatedly positioned as a key response to shorter product lifecycles, increased product variety, and more heterogeneous customer demands (Andersen et al., 2017). But how much flexibility is needed and how much to build in must be considered. In reconfigurable manufacturing, critical decisions must be made in early design stages to specify the required capability for change, for example, the degree of scalability and convertibility is treated as a design phase choice that shapes later reconfiguration (Andersen et al., 2017).

This motivation is reflected in how reconfigurable manufacturing systems are contrasted with traditional dedicated and flexible systems. Andersen et al. (2017) describe, customization, as machine and system flexibility being limited and tailored to a specific part or product family, thereby reducing the traditional efficiency and flexibility trade-off. Here, the argument is not to maximize flexibility, but to avoid over flexibility by constraining flexibility to what is relevant for the product family and expected change patterns (Andersen et al., 2017).

Above, suggests that flexibility should be treated as a bounded requirement, if boundaries are not explicit, flexibility can become a moving target that expands functionality, variants, and integration cases, leading to increasing engineering and validation burden rather than reducing it (Andersen et al., 2017).

In modular and hybrid plant contexts, flexibility is explicitly treated as a trade-off rather than a free benefit. Hybrid modular, monolithic plants are described as involving a trade-off between flexibility and optimality, and decisions such as the modularity degree enlarge the solution space and introduce additional engineering complexity (Markaj et al., 2022).

Markaj et al. (2022) conceptualize flexibility as an operational window, defined by the range of admissible operating conditions that are intentionally covered by the system's design. Building on this definition, the degree of flexibility can be operationalized as a design variable, the width of the operational variation space the system is expected to cover (Markaj et al., 2022).

From an implementation and operations perspective, flexibility is further linked to engineering effort and competence requirements. In a PnP context, the cost of flexibility is described as depending on setup cost and competence, where setup cost includes the time needed to adapt programs, plan operations, and configure hardware and software (Ramasamy et al., 2023).

3.2.3 Human usability

In this thesis, practical cloning is treated as a socio technical requirement rather than a purely technical property. Even if interfaces and integration mechanisms are standardized, the cloned solution must remain possible to install, learn, operate, and maintain in real production settings. This aligns with Industry 4.0 human factor research that frames successful system development as dependent on how humans interact with, understand, and work with the system across the lifecycle (Neumann et al., 2021). In a similar direction, the Operator 4.0 concept emphasizes that future production systems combine automation and human capabilities through adaptive support, not by removing the human role (Romero et al., 2016).

Human factors and ergonomics are positioned as necessary to optimize both human well-being and system performance, and should be treated as a design input early, rather than something evaluated after deployment (Neumann et al., 2021). This is reinforced in collaborative robotics literature, where ergonomics and human factors are discussed as a requirement for implementing safer workstations, implying that "works technically" is insufficient if interaction creates avoidable human strain or operational risk (Cardoso et al., 2021).

Recent reviews of cobot era production systems suggest a compact set of human centered dimensions that are directly usable as operational criteria, physical ergonomics, mental workload, trust, acceptance, and usability (Faccio et al., 2022). For practical cloning, these dimensions can be interpreted as repeatability conditions, each cloned instance should preserve a stable interaction logic, avoid increased cognitive supervision, and support safe and comfortable work practices. This is important because evidence syntheses highlight that research and implementation have historically focused strongly on physical safety, while mental well being and psychosocial effects, for example, stress, autonomy, satisfaction, also influence long term sustainability and acceptance of collaborative automation (Storm et al., 2022).

Acceptance and trust are particularly critical in early operationalization. Factory worker acceptance is described as multi factorial and sensitive to perceived safety, job security concerns, changes in routines, and the degree of human centered design and controllability during introduction (Baumgartner et al., 2022). Trust, in turn, is strongly influenced by robot-related factors, especially performance. meaning that unstable behavior and unreliable task execution can directly undermine willingness to rely on the system and can increase compensatory monitoring work (Hancock et al., 2011). In shared workspace collaboration, fluency is described as coordinated teamwork quality, and is positioned as relevant because acceptance can depend on how smoothly human and robot coordinate actions (Hoffman, 2019).

3.3. Chapter 3 summary

Cloneable automation is connected to plug and produce, modularity, reconfigurable manufacturing, bounded flexibility, and human usability, but it is not identical to any one of these concepts. The theoretical framework show that repeated reuse depends on modular structures, standardized and compatible interfaces, explicit variation boundaries, and mechanisms that reduce repeated integration work. The literature also shows that flexibility and reconfigurability create new complexity if the intendent design is not clearly defined. In this stud, the most important theoretical implication is that cloneability should be understood as controlled and usable reuse within defined boundaries, not as unlimited flexibility or simple copying.

4. Conceptualising cloneable automation systems

Cloneable automation is not explicitly defined in the academic literature. Related concepts exist, such as modular automation and PnP, but they do not yet provide a shared set of attributes that are both technically meaningful and practically usable across contexts. This section develops a working definition and clarifies the boundaries of what may be cloned.

4.1. Definition and boundaries

In this thesis, cloneable automation is used to describe automation solutions where both automation logic and physical configurations are designed for rapid replication across different contexts with minimal additional engineering effort. This idea aligns with reconfigurable manufacturing, where systems are designed for rapid change in structure, hardware, and software in order to adjust production capacity and functionality (Bortolini et al., 2018).
A practical boundary for what is cloned is often the module. In modular process automation, each module, also referred to as a Process Equipment Assembly (PEA), fulfills at least one process function that is encapsulated (Hoernicke et al., 2020). Cloneability at this level means that a module can be replicated as a standardized type, while exposing only the information and services needed for integration.
This creates a distinction between project independent module engineering and project-dependent plant engineering. In modular process plants, the Module Type Package (MTP) is a vendor independent description that enables the seamless integration of modules into a Process Orchestration Layer (POL) (Hoernicke et al., 2020).

4.1.1 Copy vs clone in an automation system context

Many automation projects replicate solutions by reusing existing artifacts, for example, PLC program blocks, HMI screens, configuration files, and then adapting them for a new customer or station. This practice is commonly described as clone and own, where an existing system variant is copied and modified to create a new variant (Echeverría et al., 2021; Rosiak et al., 2021). This approach is attractive because it enables fast delivery, but it becomes hard to sustain as the number of variants grows, since duplicated artifacts diverge and maintenance requires change propagation and repeated re-validation work across variants (Krüger & Berger, 2020; Rosiak et al., 2021).

In this thesis, copying is defined as one off replication by duplication and manual adaptation. It replicates artifacts, for example, code and configuration but does not replicate an engineered pathway for predictable reuse. As described for IEC 61131-3 based automated production systems, copying, pasting and modifying is used both locally, for example when cloning program units, and at system level, for example, cloning and adapting whole systems, and this non-systematic reuse does not scale due to quality and maintenance issues such as bug propagation (Rosiak et al., 2021). Similarly, Clone & Own is described as generating a new product by reusing artifacts from an existing product and adding small changes, but manual handling becomes difficult in industrial settings as product families grow and implementations become long and complex (Echeverría et al., 2021).

Cloning or cloneability, is defined here as repeatable instantiation from a solution that is engineered for reuse. This aligns with platform oriented reuse, instead of duplicating a whole solution, a reusable core is created and variants are derived through configuration using explicit variability mechanisms and models, for example, feature models and configurators (Echeverría et al., 2021; Krüger & Berger, 2020; Kuiter et al., 2018). This shifts effort from repeated re-engineering to upfront architecture and interface design, platform oriented reuse scales but requires higher initial investment, clone and own on the other hand is initially cheap and flexible but tends to impose increasing long term costs (Krüger & Berger, 2020)

Applied to automation systems, this thesis treats a solution as cloneable when replication is primarily done by configuration and controlled variability, within defined bounds, rather than by duplicating and editing implementation artifacts. To achieve this, cloneability requires that the solution's variability, including interfaces, is explicitly captured so that a new instance can be instantiated predictably, instead of being rebuilt through ad hoc modifications (Krüger & Berger, 2020; Kuiter et al., 2018)

4.2. Enabling mechanisms

Encapsulation is a primary mechanism for cloneability, because it makes functional behavior reusable without exposing internal implementation details. In modular process plants, process functions are encapsulated at the PEA level and offered as services, with integration supported through the MTP description artefact (Hoernicke et al., 2020). In discrete manufacturing, a similar idea is reflected in skill based approaches, where a unified abstraction is used to describe the skills of resources and derive executable manufacturing procedures that are exposed as services and orchestrated at runtime (Pfrommer et al., 2015).

Standardized interfaces and information models are required for a cloned resource to work in a new context with limited manual engineering. One strategy is to rely on vendor neutral descriptions of modules, topology, and behavior. Automation Markup Language (AML) is one example of a standardized storage format that can represent plant topology and behavior descriptions (Bennulf et al., 2022). AML is an XML based engineering data exchange format for plant engineering information, used to transfer structured descriptions of equipment topology, interfaces, and related artefacts across disciplines and lifecycle phases (Henßen & Schleipen, 2014; Hoernicke et al., 2020). This supports cloneable automation by enabling vendor neutral module or work cell descriptions to be reused between engineering tools and projects, reducing manual re modelling when a solution is replicated (Danny et al., 2017; Reichler et al., 2019).

For vertical communication, modular process plant modules can provide an OPC UA server interface that exposes the information required by the orchestration layer (Hoernicke et al., 2020). Such standardization is also framed as a way to tackle complexity in modular plants by using standardized interfaces for physical and information connections between PEAs (Markaj et al., 2022).

Decoupling and late binding reduce re engineering effort by separating production goals from specific resource implementations. In multi agent PnP approaches, each resource logic can be encapsulated in an agent, and agents can communicate via a standardized interface to reach manufacturing goals (Bennulf et al., 2022). A key element is that unique local behavior can be provided through configurations rather than manual programming, which supports predictable replacement or replication of resources (Bennulf et al., 2022).

Configuration management becomes critical when cloneability is pursued at scale, since replication must include parameters, versions, and reusable engineering artefacts. In agent-based PnP systems, a database can be used to store agent configurations, enabling multiple users to edit data simultaneously and supporting automated deployment (Bennulf et al., 2022). Storing configurations as standardized objects, for example in JSON or XML formats, supports the idea that a cloned resource can fetch its behavior on instantiation instead of requiring code duplication (Bennulf et al., 2022).

4.2.1 Clonability vs standardization

Standards for communication, information models, module, skill descriptions are consistently positioned as enablers for interoperability and PnP. For example, work on Industry 4.0 architectures argues that standardization around common asset descriptions and APIs supports interoperability across heterogeneous components, and skill based control approaches similarly emphasize standardized communication interfaces and data models as prerequisites for cross vendor integration (Sidorenko et al., 2023; Zimmermann et al., 2019).
The reviewed literature indicates that clonability is not the same as standard compliance. Standards can make a module connectable and discoverable, but they do not automatically make its behavior reusable, predictable, and safe when moved to a new context. In skill based settings, standardized interfaces can hide device level complexity from the user, yet the effort of encapsulating vendor specific behavior and semantics remains a major barrier, and general description schemas still need additional, context specific constraints and assumptions to be executable in practice (Pfrommer et al., 2015; Volkmann et al., 2021; Zimmermann et al., 2019).

4.2.2 Machine learning, potential future enabler

Machine learning (ML) can be noted as a potential future enabler for industrial automation applications. ML based analysis of data has the potential to improve or enable tasks such as diagnostics, predictive maintenance, and condition monitoring (Kamm et al., 2023). For commissioning, Taets et al. (2025) propose a human in the loop Bayesian optimization approach to assist operators in tuning process parameters and demonstrate reduced commissioning time in an industrial glue dispensing case(Taets et al., 2025).

4.3. Barriers and trade-offs

Cloneability involves trade-offs, particularly between flexibility and optimality. In this thesis, optimality refers to cost- and performance efficiency when a system is tailored for a specific operating point. Increasing flexibility through modularity can reduce this efficiency due to interface and integration overhead. Hybrid modular monolithic process plants are described as involving a trade-offs between flexibility and optimality, where modular solutions can sacrifice performance compared to monolithic plants optimized for a single operating point (Markaj et al., 2022). Modularity can also increase the solution space, since more configurations and interconnections must be handled, and interface standardization is partly motivated by a need to manage higher engineering and operational complexity (Markaj et al., 2022).

Safety and validation add a further barrier to predictable low effort replication. In reconfigurable manufacturing systems, assessments are often performed manually by safety engineers after each reconfiguration, which reduces the agility benefits of PnP (Koo et al., 2018). The compliance workflow can also trigger certification and recertification overhead, for

example CE marking and conformity steps under the European Machinery Directive, which provides limited flexibility for systems that change layout or function frequently (Koo et al., 2018). After reconfiguration, the system must be re examined to identify new risks and implement new safety measures where needed, delaying return to full operational capacity (Massouh et al., 2024). This is reinforced by hazards that emerge from interactions when several components are configured together, motivating system level re validation (Massouh et al., 2024). Also, safety arguments developed at design time can be invalidated by operational changes, which motivates dynamic safety assurance mechanisms that maintain safety cases as configurations evolve (Javed et al., 2021). In this thesis, safety validation and certification are treated at this barrier level, detailed conformity assessment procedures are addressed as a limitation and a future research opportunity.

4.3.1 Legacy, brownfield vs greenfield deployment

From a system design perspective, greenfield deployments allow the automation architecture, interface assumptions, and integration approach to be defined consistently from the outset, in brownfield deployments the work must go around the already installed base and migrate existing equipment. In brownfield contexts, companies have invested heavily in production equipment that is not capable of Industry 4.0 communication, and it is often not practical or economical to replace all legacy machinery (Etz et al., 2020).
A recurring constraint in brownfield settings is heterogeneity in both interfaces and capability. Legacy systems often include machines from different eras with different communication protocols, and limited instrumentation can force manual observation and adjustment of process parameters (Tran et al., 2022). At the controller level, legacy PLC programs may only support numeric types, and embedding additional OPC UA functionality directly in the PLC can create unacceptable computational side effects and timing risks for real time control (Lee & Sung, 2022).

As a result, brownfield integration is frequently achieved through wrappers, gateways, or agent layers that expose legacy data and services through vendor neutral interfaces while leaving the underlying shop floor control intact. Etz et al. (2020) describe OPC UA gateways that convert proprietary protocols and expose data via an OPC UA server while leaving lower level communication unaffected. Tran et al. (2022) similarly discuss gateway based connectivity upgrades and wrapper patterns, for example Modbus to OPC UA, to adapt a broad range of legacy machines, and Lee and Sung (2022) demonstrate an OPC UA agent running on a separate machine with shared memory synchronization to enable OPC UA information services for legacy PLCs.

These retrofit layers introduce trade-offs that matter for practical cloning. Gateway hardware requirements can increase for safety relevant applications (Etz et al., 2020), and timing interference must be investigated to ensure control constraints are satisfied when additional functions are introduced (Lee & Sung, 2022). Brownfield upgrades can also expand the cybersecurity attack surface by adding sensors, connectivity, and security retrofits that may introduce new vulnerabilities and testing challenges (Tran et al., 2022).

4.4. Working concept model: Cloneability attributes

Based on the mechanisms and boundaries above, cloneability can be treated as a set of attributes that together reduce the effort of replication while keeping the system usable in practice. Here follows attributes of interest for enabling a cloneable automation system.

1. Modularity: Operational functions are compartmentalized into independent units that can be rearranged among alternate production schemes (Bortolini et al., 2018).

2. Integrability: Modules can be connected rapidly and precisely through mechanical, informative, and control interfaces, supported by standardized interfaces between PEAs (Bortolini et al., 2018; Markaj et al., 2022).

3. Self-description: Resources provide device and localization information, and related safety information, when being connected, supporting automated identification during integration (Koo et al., 2018),

4. Service orientation: Encapsulated functionalities are exposed as triggerable services, enabling orchestration while hiding implementation details (Hoernicke et al., 2020; Pfrommer et al., 2015).

5. Interoperability: Vendor neutral models and protocols such as AutomationML and OPC UA enable consistent information exchange between modules and orchestration layers (Bennulf et al., 2022; Hoernicke et al., 2020).

6. Convertibility: The system can transform functionality to suit new production and market requirements without being rebuilt from scratch (Bortolini et al., 2018).

7. Scalability: Capacity can be modified by adding or removing resources, enabling replication by extension rather than full redesign (Bortolini et al., 2018).

8. Goal oriented planning: Runtime orchestration is based on manufacturing goals rather than fixed sequences, for example through dynamic service orchestration or agent negotiation (Bennulf et al., 2022; Pfrommer et al., 2015).

4.5. Chapter 4 summary

Cloneable automation is defined as more than copying an existing solution, the distinction between copy based reuse and cloneability is important because copying transfers artefacts, while cloneability requires an engineered structure that supports predictable reuse. The working concept models intended idea is to identify attributes that describe what makes an automation solution cloneable. These attributes are later translated into guidelines, meaning that the attributes define cloneability while the guidelines show how the attributes can be used in PLC, robot, and HMI design.

5. RESULT

This chapter presents the empirical findings from the interviews and the case related material. The result is structured around how reuse is described in current automation projects, which building blocks consume engineering and integration time, what creates surprises during commissioning, and which conditions enable or limit more repeatable reuse. The focus in this chapter is to present the interviewees' descriptions before the material is analyzed together with theory.

Note: The respondent quotes in this chapter have been freely translated from Swedish into English by the author as accurately as possible.

5.1. Recent reuse in current automation project

Across the interviews, reuse is described as present in current automation work, but most often as partial rather than complete. The interviewees describe that reuse usually starts from an earlier concept, a base program structure, a blueprint, a library, a communication block, or a mechanical sub solution

One interviewee within process automation describes that work often begins from an earlier blueprint, but that the new process still requires manual adaptation before the solution can be implemented. Another interviewee within robot automation describes a similar pattern, where new projects begin by looking back into an existing “toolbox” of concepts, fixtures, communication libraries, and HMI modules, while the final solution still becomes customer specific. A third interviewee describes that what is most often reused is a base program structure in PLC and robot programming, while the application specific part is more difficult to carry over unchanged.

A similar pattern is described in the seventh interview, focused on robot automation for material handling. The interviewee explains that they have what is referred to as a standard product, but that the delivered solution still rarely remains fully standard. What is reused more consistently is instead a base program, a recurring HMI structure, and a software centered way of designing the cell, while robot handling, grippers, layout, and customer integration still become project specific.

> *“We have a product that we call standard. But it never really becomes standard … we have a base program, the code is always ready, and the same HMI … we basically adapt the robot cell to the software.” (Automation Expert 7)*

> *“You often start from a blueprint, you could say … but there always has to be manual adaptation for the new type of automation you actually want to do.” (Automation Expert 2)*

> *“When you start sketching new automation solutions, you always go back and look at what you have in the toolbox … what have we done before that resembles this in some way?” (Automation Expert 6)*

The same pattern also appears in the more detailed examples given by the interviewees. One interviewee describes a robot cell where the ideas around a turntable, the control structure, and the robot communication could be reused, while the change of PLC brand still meant that code had to be rewritten. Another interviewee describes how reuse becomes much stronger when a block or a library is already prepared and proven in a real situation. In one example related to data collection and OEE communication, a task that previously took several hours was described as being reduced to a matter of minutes when a standardized solution already existed.

"We recently built a robot cell where we reused large parts of a previous concept with a turntable … The basic logic is reused, and much of the mechanical part, such as the turntable … we reuse that." (Automation Expert 1)

"When we have tested this way of doing this, we have gone from implementing the communication … which might take five hours, to it now taking ten minutes instead." (Automation Expert 3)

Boundaries to reuse were also mentioned during the interviews, for example, robot paths, process specific functions, project unique logic, and interfaces towards a particular customer environment were described as harder to transfer directly. One interviewee state that robot programs themselves are often written again, while parameterization is more realistic at product level, such as dimensions and other product related properties. Another interviewee described that reuse is easier at the level of base logic and structure than at the level of the full application. From the interviews, reuse was primarily described as reuse of a stable core, rather than as a direct transfer of a whole finished system.

"What is reused is a certain base program structure, for example in both PLC programs and robot programs." (Automation Expert 5)

"I would say definitely parameter-controlled … but the robot program itself … it is new every time … 10–15 percent at most." (Automation Expert 6)

Also, reuse is described as strongly parameterized and modular, much of the work is described as configuration of prepared software rather than writing entirely new code, its also described that they are not yet at a point where a robot cell can simply be instantiated without further engineering.

"You need to program very little, mostly parameterization … the goal is to reach a point where you have software where you just set which type of robot cell you have … but we are not quite there yet." (Automation Expert 7)

5.2. Building blocks that consumes engineering and integration time

A recurring theme in the interviews is that the largest cost drivers in automation projects are not only the physical components, but also the engineering time required to make the solution work. Several interviewees describe software development, programming, integration, and commissioning as the areas where many hours are consumed. This includes both initial development and the time required to test, debug, and stabilize the solution.

One interviewee describes the main cost as "man hours", meaning engineering time that the customer pays for. Another describes the work in similar terms, with many hours spent on construction and programming. In the material, this makes software and engineering effort appear as a central part of the project cost picture, especially when the solution is not based on already prepared structures.

"There are often very many engineering hours … that is the part the customer has to pay for in that situation." (Automation Expert 2)

"There are many engineering hours involved in the whole thing … there are many design hours. There is a lot of programming." (Automation Expert 5)

This picture is reinforced by the seventh interview, where the most expensive part is described not only as engineering hours in general, but specifically the time needed to make the software simple enough that the operator can use the machine without robot specialist knowledge.

"The most expensive thing by far is the hours … we want to remove the robot knowledge and create a machine where you do not need any robot knowledge … all the intelligence should be in the software itself." (Automation Expert 7)

The interviews also point to specific building blocks and activities that consume time repeatedly. Safety is described as an area that can become very time consuming, especially if it is not considered early in the project. Communication between systems, vendors, and components is also repeatedly described as difficult and expensive when protocols, platforms, or interfaces do not fit together directly. One interviewee also points to understanding an already installed solution as a major effort in itself, especially when knowledge about the old system is weak or tied to a few individuals.

"Safety is one area that can take a very, very long time … If you include the safety aspect from the very beginning of the project, it usually becomes much easier and much cheaper." (Automation Expert 1)

In robot related applications, one interviewee describes a difference between making a station perform the intended task once and making it run reliably over time. Fault handling, restart logic, recovery behavior, and overall availability are described as demanding parts of the software work. One interviewee explicitly describes this as the difference between making a robot station work and making it work every day with stable availability.

"It is one thing to get a robot station to do what it is intended to do … it is a completely different thing to get it to do it every day … and to have 90 percent availability." (Automation Expert 6)

5.3. Surprises during commissioning

The interviewees describe commissioning as one of the phases where hidden complexity becomes visible. A recurring theme in the material is that problems often appear when many systems, signals, and assumptions must function together in the real production environment. This means that even if a concept is considered as good during earlier development, the actual startup can still reveal issues related to communication, sequence behavior, material variation, tolerances, and missing standardization.

One interviewee describes commissioning as a step by step activity where many systems must be connected and checked in sequence. Another interviewee describes that startup problems often arise because people have been forced to invent their own solutions in the absence of standardized objects and programming structures. In this type of case, the same type of function may be implemented differently in different parts of the code, which makes the whole solution more difficult to understand and more sensitive to faults during startup.

"There are so incredibly many systems that have to be connected … commissioning takes so much time, and you really have to be careful and take it step by step." (Automation Expert 2)

"Much of what I have experienced when arriving at startup … is that people have had to invent their own solutions because there has not been any standardized way." (Automation Expert 3)

The interviews also shows that surprises during commissioning often come from the difference between the tested solution and the real production environment. One interviewee describes that the wrong material or unrealistic test conditions during build and test phases can create problems that only appear later. Another interviewee describes how differences in incoming material, carrier tolerances, lighting conditions, and reflections can disturb a solution that appeared to work earlier, especially in robot and vision related applications. Communication around new components is also described as a source of late problems, for example when the

person procuring a sensor and the person integrating it into the control system do not share the same understanding of how it should function.

> *"Surprises are usually related to not having worked with the right material during the time the system was being built." (Automation Expert 5)*

Commissioning surprises may also come from uncertainty in the customer's own product definition. If the product, tolerances, or practical behavior are not as stable as assumed earlier in the project, the robot cell may behave correctly from a programming perspective while still failing in the real application.

"Sometimes the customer does not fully understand their own products … the product did not quite match … and then we have some problems." (Automation Expert 7)

5.4. Legacy, documentation, and the installed base as constraints

One recurring point in the interviews is that automation projects are often carried out in existing industrial environments rather than in greenfield situations. Legacy is described as a normal condition. Companies often continue to use older systems for as long as they function, which means that new solutions must be able to fit into an already installed base of machines, control systems, fieldbuses, and local standards.

One interviewee describes this directly by stating that old systems are always present in industrial work, and that the supplier must be able to adapt to them because the customer will not replace functioning equipment without a clear reason. In the material from the interviews is also described the perspective of system lifetime, where automation systems are expected to remain for decades, while documentation and competence around them often become weaker over time.

> *"You always encounter that type of system when it comes to industry … then the company in question does not want to replace it in the first place … then you need to be able to adapt easily to the systems that exist at the company." (Automation Expert 2)*

> *"Systems that are installed are expected to last for 30 years … there is often weak competence around many of the systems that are running … and you become somewhat dependent on certain key people who possess that knowledge." (Automation Expert 1)*

The interviews also show that documentation quality becomes important in this context. Several of the interviewees describe documentation as uneven. In some cases, documentation is good and detailed, but in other cases very little is available beyond an electrical drawing or local knowledge. This means that reuse is often easier when the same person has worked with both the earlier and the later project, while reuse becomes more difficult when the original reasoning behind a solution is no longer visible. Standardized naming, comments in code, and consistent documentation are therefore described in the material as practical conditions for understanding and reusing earlier work.

> *"Documentation is usually quite poor, unfortunately. You may get an electrical drawing if you are lucky … otherwise you usually do not get much more." (Automation Expert 1)*

> *"It is mainly documentation, good documentation, and having good comments in both the program code and these mechanical solutions." (Automation Expert 5)*

Another interviewee describes connecting a robot cell to new equipment as relatively straightforward, while retrofitting older machines, with weak documentation and lower safety level, create a much heavier integration task. This supports the picture that the installed base shapes how reusable a solution can be in practice.

"Making a completely new setup, buying a new CNC machine and connecting a robot to it is no problem, but updating old things with poor documentation, raising the safety level, and everything around that is often a challenge." (Automation Expert 7)

Another part of the installed base constraint concerns competence and trust, one interviewee describes that older and newer systems often require different kinds of competence, and that younger personnel may lack experience with old systems, while experienced personnel may have stronger trust in the older way of working. The interviewee describe that people often trust what they already know and can handle, while the new solution may appear uncertain, legacy is not only a technical condition, also competence and acceptance issue must be considered.

> *"They want to continue believing in what they can manage in the old system, and the new system is unknown … then there are no younger people who know the old system anymore." (Automation Expert 4)*

5.5. Validation, safety, and certification

In the interviews safety and validation are described as a necessary parts of automation work, but also as time and effort consuming. Several interviewees describe that changes in an existing solution can lead to additional testing, risk assessment, and responsibility, and that safety work becomes especially costly if it is addressed too late. Safety is not described as a separate add on, but as something that strongly affects both engineering effort and the possibility to reuse a solution in a real setting.

One interviewee describes that if a third party makes a significant change in a machine, responsibility can shift in such a way that a new risk assessment, new CE marking, and broader responsibility for the machine follow. Another interviewee describes that a rebuilt control system needs to be validated fully, even if the mechanical part of the machine is already known. Yet another interviewee describes that validation must not only consider the changed function itself, but also the surrounding functions that lead up to it or are affected by it.

One interviewee describes that the same boundary to reuse happens when the solution is modified to do tasks it was not originally designed for. The interviewee states that when the customer later wants to run new details, new speeds, or functionality outside the original scope, new risk assessments are often required and CE related questions may reappear. Reuse is therefore described as easier when the expected variation has been considered from the beginning.

"If you are going to make new parts or new speeds that it was not designed for … then there will be new risk assessments … and it can affect the CE marking … it is difficult to take shortcuts there." (Automation Expert 7)

The interviews also describe that modular boundaries and well tested standard blocks can reduce the burden of repeated validation. One interviewee argues that if machines are built as modules with clear interfaces, a module can be replaced and the remaining work can be limited to checking that the interfaces are compatible. Another interviewee describes that pre tested and proven blocks can make the validation process more standardized and smoother.

> *"If you instead build machines as modules with clear interfaces … then you can replace a module and put in another one … and you only need to do a final validation that the interfaces are compatible with each other." (Automation Expert 1)*

> *"We know how it works … then you can almost have a standard for the validation process … you can make it smoother by using modular and standardized blocks." (Automation Expert 3)*

5.6. Enablers for faster deployment and more repeatable reuse

When the interviewees describe what makes repeated deployment easier, a number of recurring enabling factors appear, for example, standardization, modularity, documentation, parameterization, proven function blocks, HMI templates, compatible communication interfaces, realistic simulation, and an early decision to design for reuse rather than only for the first project.

One recurring point is that projects benefit from clear requirements from the customer side. One interviewee describes that vague specifications easily lead to mismatched expectations later, which in turn makes integration and commissioning more uncertain.

"The clearer you are with the customer, the better the projects usually become … if they have no requirements for us, then it becomes a bit vague, and then it may not turn out quite as intended." (Automation Expert 7)

In the same interview its also mentioned that its valuable to involving operators and maintenance personnel early, so the delivered solution becomes both technically suitable and better anchored in the customer organization.

Standardized blocks and libraries are described as a particularly concrete enabler. One interviewee gives an example where a communication solution reduced implementation time from hours to minutes. Another describes the value of function blocks that already include repeated functionality such as time supervision and alarms, since these functions then follow automatically when the block is used again in a new context.

> *"This thing with the time factor, that you can create alarms just by looking at the time … if it is implemented in the code, then suddenly you get it for all the drives you have." (Automation Expert 4)*

Standardized HMI objects, faceplates, and naming structures are described in a similar way.

The interviewees also describe parameterization as an important but demanding possibility, parameterization is described as good praxis, but it depends on already knowing what should vary and what should remain fixed. For that reason, several interviewees describe that reuse needs to be planned from the beginning. It is not described as something that can easily be added afterwards without prior thought, structure, and economic motivation.

> *"Parameterization is absolutely possible … but the question is how you know which parameters you should parameterize, so that is a very difficult question." (Automation Expert 5)*

> *"If you make the programs very, very smart from the start, then it is easy … to change a few parameters … but unfortunately, it is rarely someone who wants to pay for the extra work required." (Automation Expert 1)*

The organizational side of this is also visible in the material. One interviewee states that management needs to be clear that time should be invested already in project one in order to save time in project two. In another interview, its pointed out the need to involve the right functions early, rather than treating reuse as something handled only by the programmer or integrator, the interviews describe repeated reuse as depending both on technical design choices and on project and management decisions.

> *"It is also very much about being clear from management's side that this is something we must think through so that we save time for project number two." (Automation Expert 5)*

"I usually say from day one that we want a cross functional group and that we want to have these people on board early." (Automation Expert 5)

In one interview, the interviewee describes reuse as enabled by modular software building blocks, almost like Lego, where functions such as conveyors, vision, and marking are already prepared and activated depending on the application, reuse is not only a question of writing generic code, but also of maintaining a shared modular structure between projects.

"I usually compare it a little to Lego … basically just activating certain modules in the software." (Automation Expert 7)

One interviewee places strong emphasis on digital models and simulation as a way to reduce later surprises. The wish is not only to simulate the normal process, but also to provoke faults already in the digital environment so that fault handling and recovery can be developed earlier.

"If we could reach a point where we could start provoking faults in the simulation … then we would save an incredible amount of time." (Automation Expert 6)

A similar idea is also described by another of the interviewee's , where the stated aim is to complete as much testing and simulation as possible before the solution reaches the customer site. The purpose is to minimize downtime and shorten on site commissioning by shipping a solution that is already almost finished.

"We want to be 99.9 percent finished here before we send it … that is why we want to test and simulate as much as we can here before we send it." (Automation Expert 7)

5.7. Human usability and perceived complexity

Regarding human usability and perceived complexity the material from the interviews described usability as an important practical condition for repeated reuse. A reused solution is not only expected to function technically, but also to be understandable for operators and maintenance personnel. Several interviewees describe the value of familiar HMIs, recognizable alarm structures, similar navigation, and documentation that is adapted to different user groups.

One interviewee describes that operators should recognize where alarms are located and how the interface is organized when moving between machines. Another interviewee describes that HMI templates and standardized objects create recognition for both operators and maintenance personnel. A third interviewee formulates good usability in a direct way by stating that the interface should be self explanatory.

Yet another interviewee further described usability by distinguishing between operator and maintenance needs. The interviewee describes that operators often need alarms and guidance presented in a more direct way, while maintenance personnel want more depth and the possibility to go deeper down and have access to more functions in the HMI.

"Operators want alarms in a completely different way … maintenance often wants to dig down and click around until they find what they want … if you make something in between, it can turn out badly for both sides."
(Automation Expert 7)

"Then you have succeeded with the user interface … it should be self-explanatory."
(Automation Expert 5)

The material also describes usability in relation to practical interaction with the system. One interviewee mentioned that physical interaction can in some situations be better than a touch

screen, depending on environment and task. Another interviewee described that when an operator repeatedly has to use the teach pendant in a robot cell, it can be a sign that the solution has not been sufficiently adapted to the intended user.

"If they have to grab the teach pendant for a robot cell, for example, then you have probably failed a little in the programming toward the user." (Automation Expert 5)

Yet another interviewee describes the use of 3D models in the interface as a way to make faults more visual and easier to understand.

The human side also appears in relation to workarounds, safety, and cybersecurity. The interviewees describe that when new equipment, security routines, or safety functions are experienced as too slow or too impractical, people may try to bypass them. The responsibility was described as shared between the organization and the machine builder. The organization needs to provide workable routines and security culture, while the solution itself needs to be designed so that guards, interfaces, and work procedures function in daily work.

One interview described workarounds as a normal development, over time as the operators become more familiar with the machine they usually find quicker ways of working. The interviewee also places responsibility on the end customer to continue performing risk assessments after delivery.

"The end customer has an obligation to perform risk assessments regularly … it is very common for operators to find clever ways of working … if you see a broom handle or something like that by a fence, then you know something suspicious is going on." (Automation Expert 7)

> *"The human factor is the hardest thing to protect against … otherwise you find other types of solutions instead, because you want to get started." (Automation Expert 2)*

5.8. Interviewees' descriptions of cloneable automation

The last question in the interview was what a cloneable automation solution could mean, the descriptions in the material move in a similar direction, even if they are expressed somewhat differently. Several interviewees describe it as something that should be possible to copy more directly than is common today. Others describe it as a stable base, a reusable core, or a solution where only the outer layers need adaptation while the main structure can remain.

One interviewee describes a cloneable solution simply as something that can become a copy. Another describes it as something that should in principle be possible to copy straight off. A third interviewee connects it to platform thinking, where a stable ground is reused and only the project specific part changes. In the robot related material, cloneability is also connected to situations where a similar station needs to be built again in order to increase capacity, while the same material also points out that this is easier within narrow and repeated application areas than across completely different industries and contexts.

Yet another interviewee describes cloneability in a similar, but more software centered way, hardware can vary between projects, while the same software base is reused. Cloneability is not described as strict physical, but as reuse of a stable software core across similar applications.

"The hardware is replaceable, while we always use the same software base."

(Automation Expert 7)

> *"Cloneable … then I had something that could become a copy." (Automation Expert 5)*

5.9. Second site visit, code ownership, documentation, and standard based reuse

The second site visit supports the result that cloneability is closely connected to ownership of code and knowledge. A recurring point in the material was that automation code should be treated as a company asset, not as something that belongs to one developer, one supplier, or one local project. If the logic, naming, comments, and design reasoning are not documented, the machine may still run, but the company becomes dependent on people who remember why the solution was made in a certain way. In this sense, weak documentation creates a future reverse engineering problem and makes reuse less stable over time.

Documentation was also described as part of the automation architecture rather than as a separate handover activity. The material from the second site visit points to an agreed company language for comments and documentation, comments that explain purpose and assumptions, and documentation that connects drawings, I/O mapping, code, hardware choices, software options, licenses, and special configurations. Documentation should preserve design intent, not only implementation details. The discussions during the second site visit also showed that documentation has to be a mandatory part of the delivery workflow, because project pressure otherwise pushes it aside.

The site visit also gave concrete examples of how repeated engineering effort can be reduced through reusable blocks and standards. Standard function blocks can be used for recurring tasks ,for example, light tower logic, bit and value conversion, alarm translation, robot startup, job data handling, and PLC robot handshakes. The value of such blocks is not only that they reduce the amount of code, but that they make the interface, tag mapping, example implementation, troubleshooting, and later updates more consistent. Each reusable block should be supported by a practical user guide next to the library file showing the intendent idea of the block and examples how it can work.

For robotics, the second site visit highlighted the need for a recognizable company standard. Robot tags, sequences, data structures, startup routines, job data handling, and I/O mapping should follow the same logic across robots and sites where possible. The specific naming convention should be decided and made mandatory, but the important point for cloneability is not a particular prefix or letter. The important point is that the active step, missing condition, alarm, and sequence meaning can be understood without reading hundreds of lines of local code.

The second site visit also supported the idea of a stable core with local variation. The core should remain recognizable across sites and machines, while expected variation is handled through parameters, recipes, documented options, and platform specific variants where needed. This means that standardization should not make the solution rigid. Instead, standardization should make local variation safer by keeping the reusable core stable and moving expected differences into controlled configuration.

“It is not about building the best code, it is about building the most usable code.” (Automation Expert, site visit 2 discussion)

“The standard exists in the core, and then you build something else on top of it.” (Automation Expert, site visit 2 discussion)

5.10. Chapter 5 summary

The empirical result shows that reuse already exists in current automation work, but often as partial reuse of base programs, blueprints, blocks, HMI structures, and earlier project knowledge. The most time consuming work appears in the engineering intensive connections between PLC logic, robot behavior, communication, HMI, safety, validation, commissioning, and the installed base. The second site visit supports this result by showing that cloneability also depends on company ownership of code, documentation as a release gate, reusable standard blocks, robot standards, and a stable core that can support local variation. Overall, the result indicates that practical cloneability depends on both technical structure and organizational ability to document, teach, maintain, and reuse the standard.

6. ANALYSIS AND DISCUSSION

This chapter combines the theoretical framework and the empirical findings in order to answer the research questions. The analysis discusses cloneability as engineered reuse rather than copy based reuse, identifies where engineering effort accumulates in practice, and refines the proposed cloneability attributes. The chapter also translates the findings into tentative design guidelines for PLC, robotics, and HMI.

6.1. Cloneability as engineered reuse rather than copy based reuse

The empirical material supports the theoretical distinction between copy based reuse and cloneability, also it shows that current industrial practice is often located somewhere in between these two positions. In the theoretical framework, copy or clone and own is described as duplication followed by manual adaptation, while cloneability is closer to platform oriented reuse, where a stable core is instantiated through controlled variability and configuration rather than rebuilt through ad hoc editing (Echeverría et al., 2021; Krüger & Berger, 2020; Kuiter et al., 2018). The interviews point in the same direction. Reuse is common, but most often as reuse of a blueprint, a base program, a library, a communication package, or a recurring HMI structure. One finding in the material is that the final solution still becomes customer specific and requires manual engineering around the reused core. This means that present reuse practice is stronger than simple copy paste work, but usually not yet equal to full cloneability.
The interviews gave a new view of what should be understood as cloneable in an automation context. The common description is not that a whole finished system can be copied straight off, instead the common direction is a stable and reusable core where mainly the outer layers need adaptation. This aligns with the theoretical argument that cloneability should be built around explicit variability mechanisms and bounded reuse, rather than around duplication of entire artefacts (Krüger & Berger, 2020; Kuiter et al., 2018). It also aligns with reconfigurable manufacturing literature, where the important design question is not to maximize flexibility in general, but to define what kind of change the system is intended to cover from the beginning (Andersen et al., 2017; Markaj et al., 2022). In the empirical material, this boundary becomes visible in the repeated distinction between what can be reused as a core and what still becomes project specific, for example, robot paths, customer interfaces, process details, and adaptation to the installed base. This also shows that cloneability in this thesis should be understood as a degree rather than as a binary state, for example, a solution may be more or less cloneable depending on how much of the engineering work can be moved from late project adaptation to earlier architecture, interface, and parameter design. The empirical material also adds an important practical condition that is less explicit in the purely technical literature, a solution is not sufficiently cloneable if it can only be copied technically but still requires specialist knowledge for ordinary operation, maintenance, or adaptation. Here, human usability becomes part of the definition, not only a side issue. This supports the socio technical perspective in the theoretical framework, where successful automation depends not only on technical integration but also on how the system can be understood and used by real actors across the lifecycle (Neumann et al., 2021; Faccio et al., 2022).

The analysis shows that reuse in automation projects is often stronger than simple copy paste work, but still weaker than full cloneability. In current practice, reuse often starts from a previous project, a base program, a blueprint, a library, or a recurring HMI structure, but, the reused solution still needs manual adaptation to fit the new customer, process, interface, safety situation, or installed base. This means that reuse often becomes copy based reuse, where earlier artefacts are duplicated and then changed for the next project.

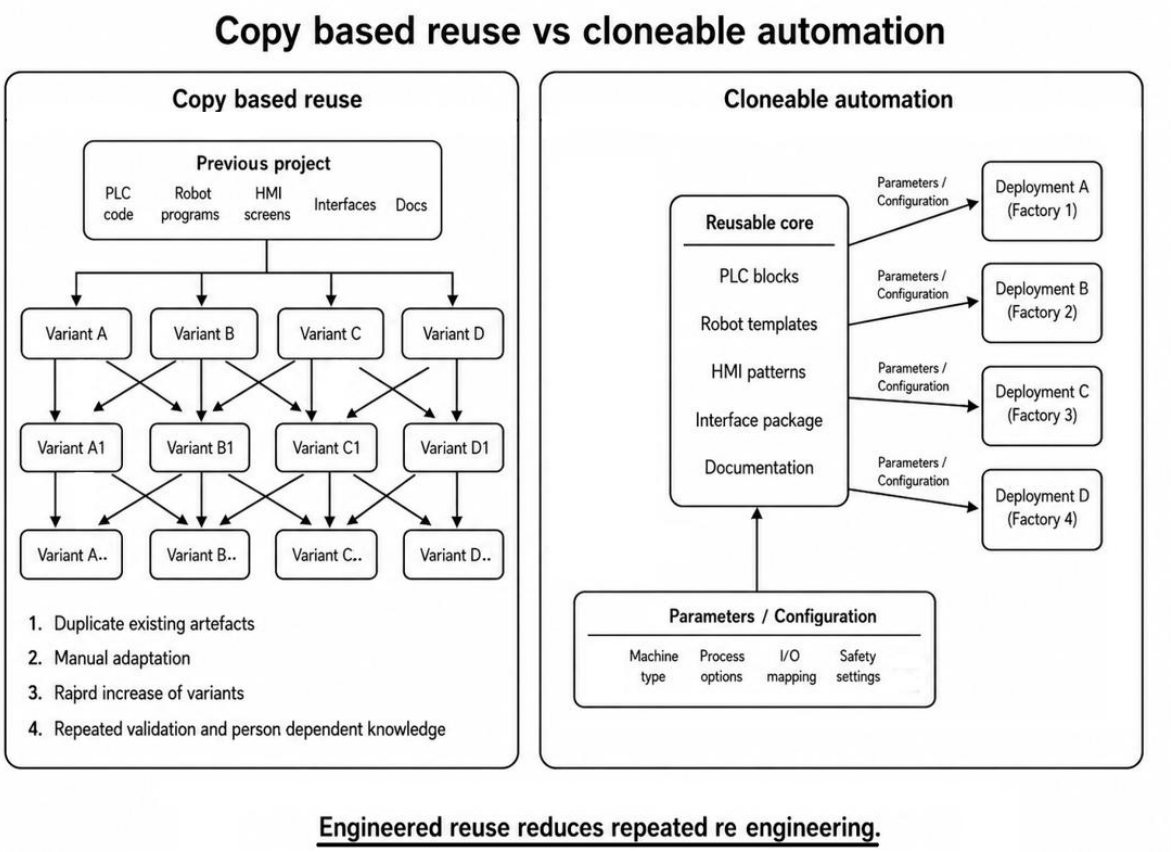


Figure 2: Copy based reuse vs cloneable automation. Figure created by the author.

Figure 2 illustrates the difference between copy based reuse and cloneable automation. In copy based reuse, PLC code, robot programs, HMI screens, interfaces, and documentation are copied from an earlier project and manually adapted. This can reduce effort in the beginning, but it also creates many variants over time. Each variant may need new adaptation, new validation, and knowledge from the person who made the original changes.

Cloneable automation instead builds reuse around a stable reusable core. The core contains reusable PLC blocks, robot templates, HMI patterns, interface packages, and documentation. The variation between deployments is handled through parameters and configuration rather than through repeated manual re engineering. This means that cloneability is not only about copying an earlier solution, but about designing the solution so that it can be reused in a controlled way.

Based on above, RQ1 can be answered as follows:
A cloneable automation system is characterized by a stable reusable core, modular boundaries, standardized interfaces, bounded parameterization, strong documentation, and preserved usability across repeated deployments, and it differs from traditional replication by reducing dependence on manual re engineering of duplicated artefacts.

6.2. Where engineering effort accumulates in practice

The main engineering cost drivers are concentrated less in isolated hardware items and more in the work needed to make different parts function together reliably in one real context. The theoretical framework already points to setup cost, competence requirements, integrability, and reconfiguration overhead as central issues when flexibility is pursued(Bortolini et al., 2018; Markaj et al., 2022; Ramasamy et al., 2023). The interviews confirm this, but with a more

practical emphasis. Software development, integration, commissioning, safety work, and understanding older systems are repeatedly described as the most time consuming parts. This means that the most important building blocks are not only physical resources, but the engineering intensive interfaces between PLC logic, robot behavior, HMI, sensors, safety functions, and the surrounding customer environment.

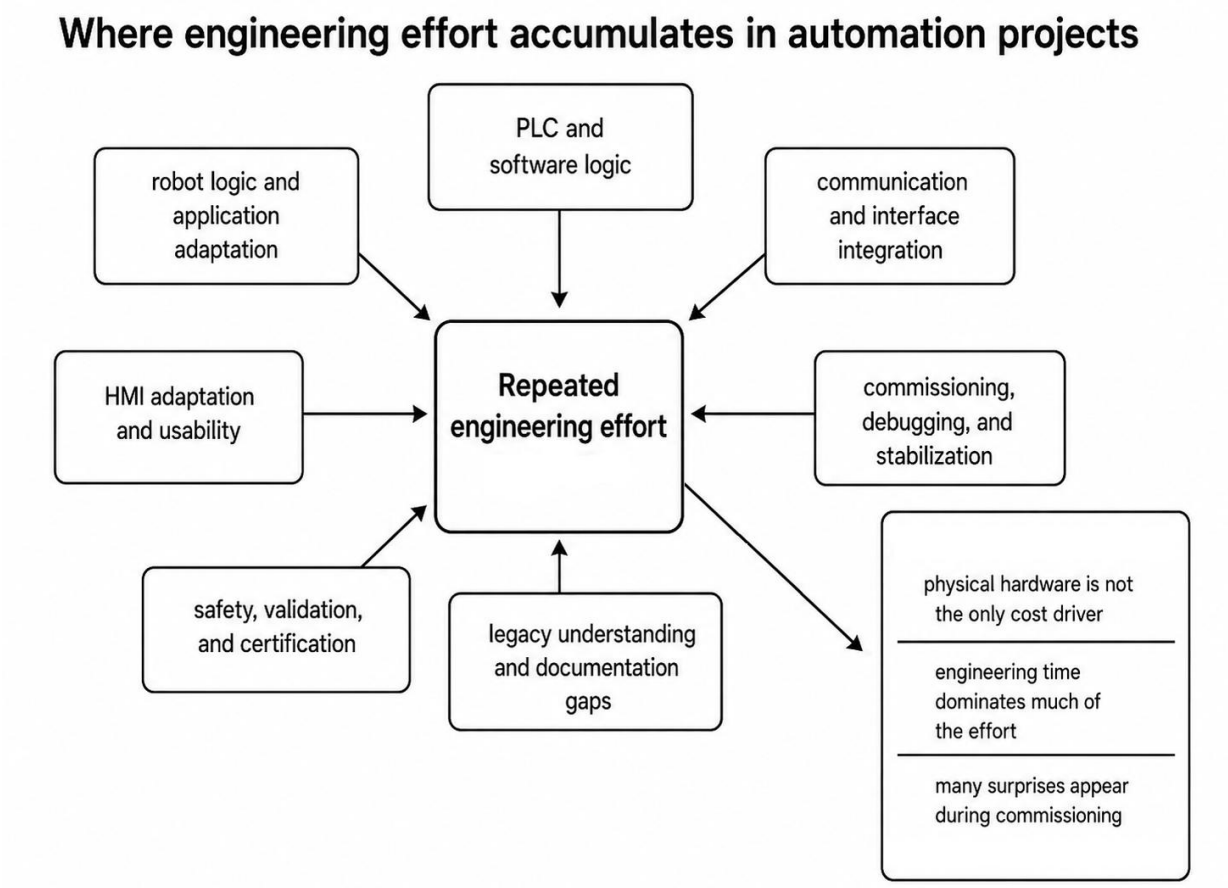


Figure 3: Main areas where repeated engineering effort accumulates in automation projects. Figure created by the author.

Figure 3 summarizes where repeated engineering effort accumulates in practice. PLC and software logic, robot logic, communication interfaces, HMI adaptation, safety validation, legacy understanding, and commissioning all contribute to the total effort. These areas are connected because a change in one part of the system often affects other parts. For example, a change in robot logic can affect PLC sequences, HMI alarms, safety validation, and commissioning work.

The figure supports the answer to RQ2, where the most time consuming building blocks are not only separate components, but the engineering intensive connections between PLC, robot, HMI, safety functions, communication interfaces, and the surrounding production environment. This means that cloneability must reduce repeated engineering work across the whole automation structure, not only reuse individual code blocks or hardware modules.

Commissioning becomes especially important because it acts as a practical stress test to see if the intended reuse logic actually holds. In theory, modularity and interoperability should reduce re engineering effort by enabling more predictable integration (Hoernicke et al., 2020;

Pfrommer et al., 2015). In the empirical material, commissioning repeatedly reveals hidden assumptions that were not fully captured earlier, for example, material variation, tolerances, lighting conditions, communication mismatches, sequence dependencies, and missing standardization. Here, cloneability should not be judged by, if a solution can be assembled or programmed once, but if it can be made robust with limited additional engineering when transferred to a new but related context. This helps explain why fault handling, restart logic, recovery behavior, and stable availability were described as so demanding in robot related applications. A solution that only works under nominal conditions is reusable in a weak sense, but not strongly cloneable.

The analysis also shows that legacy and safety are central conditions that shape practical cloneability. Brownfield literature describes heterogeneity, wrappers, and retrofit constraints as major barriers to plug and produce style deployment (Etz et al., 2020; Tran et al., 2022). The interviews support this strongly. Reuse becomes more difficult when older equipment has weak documentation, lower safety maturity, and competence tied to a few key individuals.

The same pattern appears for safety and validation. The theoretical framework treats safety as a recurring barrier to reconfiguration because changes can trigger new risk assessments and recertification work (Koo et al., 2018; Massouh et al., 2024). The empirical material makes this highly concrete by that later changes, unclear initial requirements, and variation outside the original design envelope create repeated validation effort.

With above support, RQ2 should not be answered only as a list of expensive components. The most time consuming building blocks are PLC and software logic, robot application logic, communication and interface solutions, HMI adaptation, commissioning, and validation work, especially where they must be fitted into an existing installed base.

6.3. Refined cloneability attributes from the combined analysis

The working concept model in the theoretical framework proposed modularity, integrability, self description, service orientation, interoperability, convertibility, scalability, and goal oriented planning as clonability attributes. The empirical material supports several of these, especially modularity, integrability, interoperability, and bounded convertibility. But, the interviews show that these attributes only become practically valuable when they are translated into more operational conditions such as standard blocks, prepared interfaces, parameterization, and realistic verification. The empirical material does not reject the theoretical attribute model, but makes it more concrete and more limited to the maturity level visible in current industrial work.

One refinement concerns documentation. In the theoretical framework, documentation is implied in concepts such as self description, interoperability, and modular interfaces, but the interviews show that documentation needs to be treated as an explicit cloneability attribute in its own right. Without comments, naming structures, parameter lists, interface descriptions, and preserved design reasoning, reuse becomes dependent on the original engineer making it weak over time. Documentation is what allows a modular or standardized solution to remain understandable after the first project. The empirical material adds a time dimension to cloneability, it is not enough that a solution is reusable when it is fresh in the mind of the developer, it should remain reusable when knowledge is transferred between people and across years.

Another refinement concerns parameterization and bounded variation. The theoretical framework argues that flexibility should be constrained to the variation the system is intentionally designed to cover (Andersen et al., 2017; Markaj et al., 2022). The interviews strongly support this. Parameterization is described as valuable, but only when the expected variation is understood already from the beginning, otherwise the generic solution easily becomes too vague, too difficult to validate, or too costly to build. This means that parameterization should not be treated as a general good in itself, but as a mechanism for controlled variability within a defined scope. This is an important analytical bridge between the literature on flexibility and the empirical descriptions of why later customer changes often create disproportional engineering and safety work.

Yet another refinement concerns verification readiness. The empirical material repeatedly highlights proven blocks, realistic simulation, and pre tested communication solutions as enablers for faster deployment. This can be interpreted as a practical extension of integrability and service orientation. The important issue is not only that functions are modular, but that they are sufficiently proven to reduce uncertainty during the next instantiation. Cloneability depends on whether the reusable core carries with it not only code and interface structure, but also previously validated behaviour, fault logic, and predictable interaction patterns. This is especially important in robotics, where the interviews showed that stable daily operation is a more demanding goal than making the cell perform one nominal cycle.

Yet another refinement concerns human usability and organizational anchoring. The theoretical framework already positions cloneability as a socio technical issue (Neumann et al., 2021; Romero et al., 2016). The interviews strengthen this clearly. Familiar HMI layouts, recognizable alarm structures, role adapted views, and early involvement of operators and maintenance personnel were described as practical conditions for successful reuse. Management support and cross functional involvement were also described as necessary, because repeated reuse requires investment already in the first project. The analysis suggests that cloneability is not only a technical property of an artefact, but also a capability that depends on whether the organization is willing and able to maintain standards, documentation, and reuse discipline over time.

Also, some of the more advanced mechanisms in the theoretical framework, for example, self description in a stronger plug and produce sense and goal oriented runtime orchestration, were less visible in the empirical material. This does not make them irrelevant, but perhaps they belong more to a future or higher maturity state than to the core of cloneability in the studied context. The interviews point more strongly towards a pragmatic path based on prepared libraries, stable software cores, compatible interfaces, and bounded parameterized variation.

Based on the combined literature and empirical analysis, RQ3 can be answered by distinguishing between attributes that define cloneability and attributes that enable cloneability. Cloneability is defined by a stable reusable core, modular boundaries, standardized and compatible interfaces, bounded parameterization, and preserved usability across repeated deployments. These attributes make it possible to replicate an automation solution without relying mainly on manual re engineering of copied artefacts. Cloneability is enabled by strong documentation, proven reusable blocks, realistic verification, safety and validation readiness, and early organizational anchoring. Together, these attributes show that cloneability is not only a technical property, but also depends on how well the solution can be understood, validated, maintained, and reused by the organization.

Based on the combined literature and empirical analysis, cloneability can be understood through both defining characteristics and enabling conditions. The defining characteristics describe what a cloneable automation solution should contain. The enabling conditions describe what must be in place for the solution to remain reusable, understandable, and valid over time.

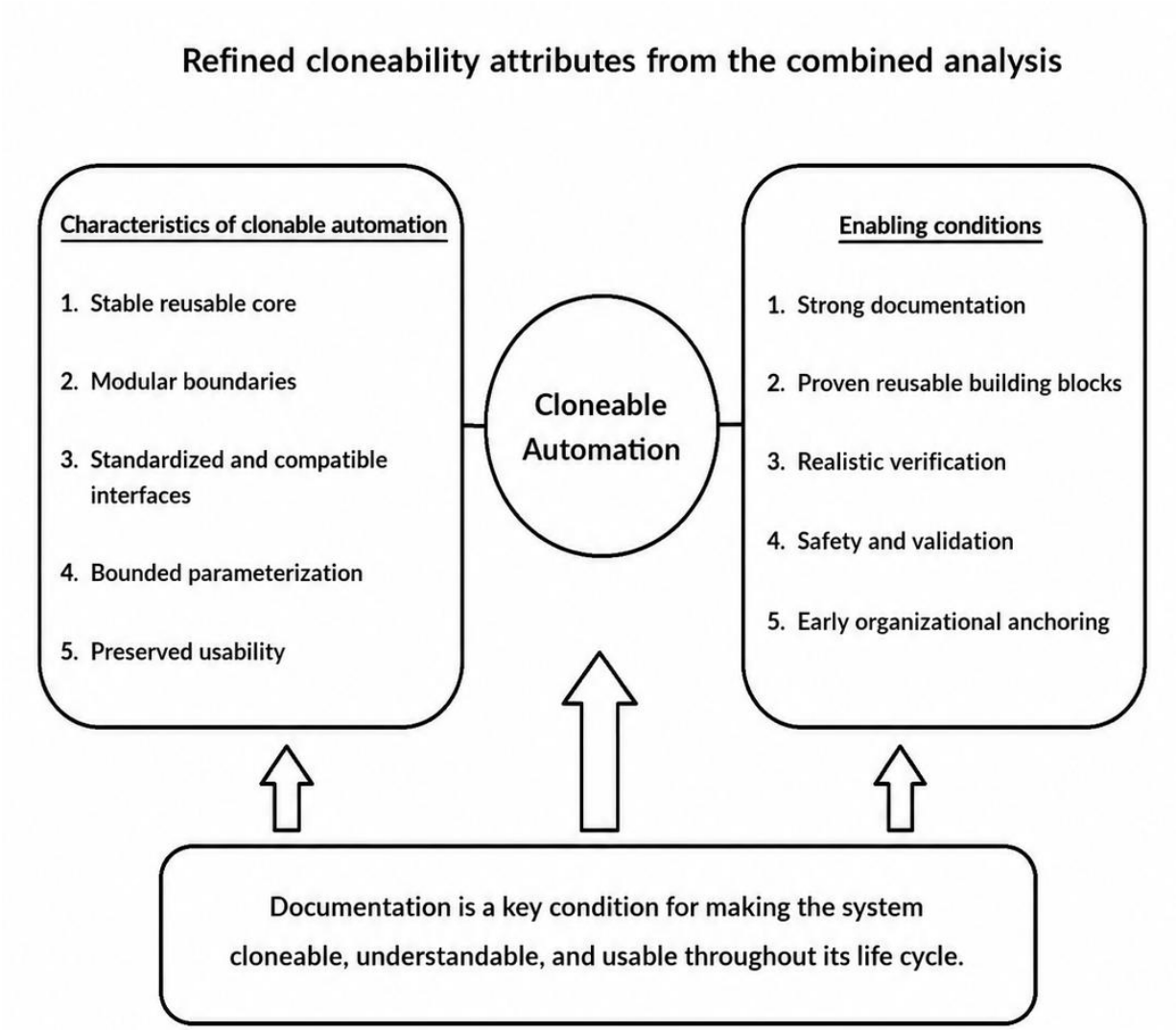


Figure 4: Refined cloneability attributes and enabling conditions from the combined analysis. Figure created by the author.

Figure 4 presents the refined cloneability attributes from the combined analysis. The characteristics on the left side are a stable reusable core, modular boundaries, standardized and compatible interfaces, bounded parameterization, and preserved usability. These attributes describe the structure of a cloneable automation solution. They show that cloneability depends on a core that can be reused across deployments while the expected variation is handled in a controlled way.

The enabling conditions on the right side are strong documentation, proven reusable building blocks, realistic verification, safety and validation, and early organizational anchoring. These conditions are important because a technically reusable solution may still fail in practice if it is difficult to understand, difficult to validate, or dependent on individual knowledge.

Documentation is therefore shown as a key condition. It supports cloneability by making the system understandable and usable throughout its life cycle.

6.4. Tentative design guidelines for PLC, robotics, and HMI

Based on the combined analysis, RQ4 is answered here not through a final universal architecture, but through tentative design guidelines for the building blocks where repeated engineering effort was most clearly concentrated. The value of the thesis is mainly in the how, how cloneability can applied to PLC, robotics, and HMI, under brownfield conditions and within a realistic industrial level of maturity.

The relationship between attributes and guidelines can be clarified in the following way. Attributes describe what defines cloneability and what conditions make cloneability possible. Guidelines describe how those attributes can be turned into design decisions in PLC, robot, and HMI work. For example, the attribute stable reusable core becomes a guideline to use proven PLC blocks, robot data structures, HMI patterns, and documented interfaces. The attribute bounded parameterization becomes a guideline to handle expected variation through parameters, recipes, and clearly delimited configuration instead of new code for every local case.

6.4.1 Documentation as a cross cutting requirement

Across all three areas, documentation should be treated as part of the architecture and not as a final handover activity. Based on the analysis, interface definitions, naming structures, parameter lists, state logic, alarm principles, safety assumptions, and recovery sequences should be documented in a consistent way from the first project. This is important because much of the empirical difficulty around reuse was connected to weak traceability and person dependent knowledge. Strong documentation is what allows modularity, standardization, and parameterization to survive over time, especially in brownfield environments where future engineers may need to connect new functionality to old equipment with limited local knowledge.

The second site visit made this requirement more concrete by showing that documentation should function as a release gate and a user guide, not only as a passive archive. A cloneable standard should describe the purpose of the block or structure, the interface, example use, tag mapping, troubleshooting logic, version information, and the assumptions behind the design. This is what makes the company owner of the automation knowledge, instead of depending on the person who first wrote the code.

6.4.2 PLC

For PLC design, based on the analysis, cloneability should be built around a modular and proven software core. Repeated functions should be encapsulated in standardized function blocks with stable interfaces, while project specific variation should, as far as possible, be moved into parameters, recipes, and clearly delimited I/O mappings. Alarm handling, time supervision, state transitions, and communication patterns should follow the same logic across projects, because these recurring structures were described in the empirical material as major time savers when they were already prepared. Also, the analysis shows that parameterization should remain bounded. The PLC solution should be designed for expected variation, not for unlimited generality, since too broad flexibility quickly becomes difficult to understand, validate, and maintain.

Operationally, this means that recurring PLC functions should be placed in company controlled blocks when the use case is repeated often enough. Examples from the second site visit include light tower handling, bit and value conversion, alarm translation, and clear mapping between raw machine information and HMI ready diagnostics. Reusable block can affect many later, to handle this, version control was also discussed as an important part as it enables rollback possibilities, and a controlled release process, because a reusable block can affect many later machines.

6.4.3 Robotics

For robotics, based on the analysis, cloneability is more realistic when pursued through reusable modules, task templates, signal packages, and prepared sensor libraries rather than through an expectation that full robot programs can simply be copied between projects. The empirical material shows that product handling, paths, grippers, and process details often remain application specific, but that the surrounding software structure can still be strongly reused. A cloneable robot solution should have a stable communication and control core towards PLC and HMI, while the product specific motion logic is adapted within a bounded template. The analysis also shows that realistic simulation is critical. Simulation should not only cover the nominal cycle, but also material variation, tolerances, lighting, disturbances, restart situations, and fault recovery, because many of the costly surprises in commissioning were described as appearing exactly in these areas.

For robot design, the second site visit adds that the robot standard should include startup logic, job data handling, handshake sequences, robot data blocks, I/O mapping, and one mandatory naming convention. The exact naming convention must be decided by the company, but it should be stable enough that several robots can be understood through the same structure. The guideline can help to make robot reuse recognizable and teachable, not only technically reusable.

6.4.4 HMI

For HMI design, the analysis points towards standardization of interaction rather than only standardization of graphics. Common navigation principles, familiar alarm structures, recurring object behavior, and stable naming should be kept across cloned solutions so that operators and maintenance personnel can recognize the system quickly. The interviews also show that one interface should not try to be optimal for everyone at the same time. Cloneability benefits from role adapted views, where operators receive direct and self explanatory guidance, while maintenance personnel can access deeper diagnostic and adjustment functions. Another important implication is that ordinary work should be possible without unnecessary dependence on specialist tools. If operators repeatedly need to enter robot specific tools such as the teach pendant for normal work, cloneability is weak from a user perspective even if the underlying automation logic is reusable.

The second site visit also support the HMI guideline by connecting usability to traceability across the whole automation standard. The same terminology should be used across PLC, HMI, drawings, robot standards, and documentation. The active step, missing condition, and alarm should be visible without requiring the user to read large amounts of code. In this way, HMI standardization supports cloneability by making the reusable structure visible to operators and maintenance personnel.

6.4.5 Stepwise interpretation

The guidelines above can be read as a tentative stepwise logic for cloneable automation.

1. Define the clone boundary and what variation should be supported.
2. Build modular boundaries and stable interfaces between PLC, robot, HMI, and surrounding equipment.
3. Encapsulate recurring functionality in proven blocks, templates, and parameterized structures.
4. Verify the solution under realistic operating and fault conditions
5. Including safety consequences of variation.
6. Document the core in a way that supports later transfer between people and projects.
7. Involve operators, maintenance, safety competence, and management early so that the solution becomes both technically reusable and practically workable.

Documentation, versioning, and release control should be treated as part of the reusable architecture, not as final administration.

Write the standard as a practical user guide and train users so the standard is understandable across competence levels.

This stepwise interpretation should be seen as a tentative contribution from this thesis, not as a final universal method.

6.5. Broader industrial implications

These findings suggest that cloneable automation can support industry by shifting reuse from ad hoc copying to structured configuration within predefined boundaries. For manufacturers that repeat similar automation tasks across products, sites, cells, or customer projects, a stable reusable core can reduce repeated manual engineering and make transfer less dependent on the memory and availability of individual developers. The industrial value is not only shorter programming time, but also a more reliable way of carrying knowledge, interface logic, HMI patterns, validation assumptions, and documentation from one deployment to the next. This is relevant for both larger organizations, where standards must travel across departments and sites, and smaller organizations, where limited specialist competence makes understandable and reusable solutions especially important.

The proposed model should not be interpreted as a promise that automation can be copied freely between unrelated contexts. Its value depends on maturity and discipline, the reusable core must be stable, the expected variation must be bounded, interfaces and parameters must be documented, and the solution must be verified under realistic operating and fault conditions.

Safety, commissioning, user acceptance, cybersecurity, and legacy integration are not later add ons, they are conditions that determine if reuse remains controlled. Cloneability here has a limit

when the new context differs beyond the original intended design, or when the organization cannot maintain the standards and documentation that the model requires.

For industrial practice, the model can be used as a decision support tool. Before a company invests in a cloneable solution, it should identify which parts are repeated often enough to justify a reusable core, where manual engineering accumulates, and which boundaries are safe and useful to parameterize. That evaluation helps distinguish between functions worth standardizing and functions that should remain project specific. The proposed model contributes by making reuse more explicit, it helps companies discuss not only what can be reused, but under which technical, human, and organizational conditions reuse should be attempted.

7. CONCLUSIONS AND RECOMMENDATIONS

This chapter summarizes the main conclusions from the thesis and relates them back to the aim and research questions. It presents what cloneable automation means in the studied context, which attributes and prerequisites support it, and how the concept may be used in future work. The chapter ends by pointing out areas where further research is needed.

7.1. Main conclusion

The analysis shows that cloneable automation in the studied context is less about fully autonomous plug and produce and more about reducing repeated engineering through a stable, documented, bounded, and usable core. The theoretical framework and the empirical findings largely point in the same direction, but the empirical material shifts the emphasis towards the industrial conditions that make cloneability workable in practice. The contribution of the thesis is a more practical interpretation of cloneable automation and a set of tentative design guidelines for the building blocks where repeated engineering effort was shown to be most concentrated.

This is also supported by the second site visit, showing that cloneability is also about ownership and continuity. A solution is more cloneable when the company can understand, maintain, troubleshoot, reuse, and safely improve the code without depending on one person or one supplier. This means that documentation, naming, versioning, reusable blocks, and teachable standards are not support activities outside the concept. They are conditions that allow the reusable core to survive after the first implementation.

The aim of this thesis was to conceptualize cloneable automation and identify which attributes and prerequisites can make an automation solution possible to replicate across contexts with limited re engineering, while still remaining usable for the people who are expected to work with it. The study started from the problem that cloneable automation is discussed as an interesting idea, but still lacks a clear definition and a shared set of attributes. Based on the literature study and the empirical material, cloneable automation should not mainly be understood as copying an existing solution and adapting it again, but more as a form of engineered reuse. In this study, a stable and reusable core, together with modular boundaries, standardized and compatible interfaces, bounded parameterization, good documentation, and preserved usability was identified as enablers for reusability.

7.2. Main findings

Another conclusion is that the main engineering burden in automation projects lies less in the physical hardware itself and more in the work needed to make the solution function reliably in its real context, For example, PLC and software logic, robot related logic, communication and interface solutions, HMI adaptation, commissioning, safety work, and validation are the parts that consume much of the engineering and integration time. Commissioning is especially important here, since it often reveals hidden assumptions related to material variation, tolerances, restart logic, disturbances, and the installed base. So, cloneability is closely connected to how much of this repeated effort can be reduced through a more prepared and reusable solution structure.

Yet another conclusion is that present industrial practice already contains elements of cloneability, but mainly in a partial form. The interviews show that reuse often exists in the form of base programs, software libraries, communication blocks, HMI templates, and recurring concepts, but, the final solution still often becomes customer specific and requires manual adaptation.

7.3. Contributions and limitations

This study has achieved a clearer conceptualization of cloneable automation and identified a set of practically relevant attributes together with tentative design guidelines for PLC, robotics, and HMI. The study has not verified a full universal architecture for cloneable automation, and it has not measured exact time or cost savings quantitatively. The conclusions should be understood as analytically grounded, but still tentative. More advanced enablers, for example, stronger self description, goal oriented runtime orchestration, and machine learning, were identified as interesting future possibilities, but they were not verified strongly enough in this study to be treated as the practical core of cloneability.

The study has a single case orientation, but the principles identified in the thesis are not only relevant to the studied company. The problem of repeated engineering effort, weak documentation, manual adaptation, commissioning uncertainty, and dependence on specialist knowledge is not unique to one industrial context. The findings can be understood as analytically generalizable to other assembly and automation contexts where similar reuse, integration, and usability challenges appear. The exact technical solution will still depend on the local production system, installed base, and organizational maturity, but the underlying need for a stable, documented, bounded, and usable core is relevant beyond the specific case studied in this thesis.

7.4. Practical recommendations

From a practical perspective, it is recommended that companies should define early what is expected to remain stable between projects and what variation should be allowed. It also recommend that standardization efforts should be focused on the parts where engineering effort accumulates most strongly, for example recurring PLC functions, communication structures, HMI solutions, alarm handling, and documentation routines. Another important implication is that documentation and realistic verification should be treated as part of the reusable architecture, not as something added at the end. The study also recommend that operators, maintenance personnel, and safety competence should be involved early, since practical cloneability depends on usability and not only on technical function.

Based on the second site visit, it is also recommended that companies treat documentation as a mandatory release requirement. Each reusable PLC or robot block should have a practical user guide, example implementation, interface description, tag mapping, troubleshooting information, and version information. The standard should be distributed, explained, and supported by training and management commitment, otherwise the standard risks becoming a document that exists but is not used.

7.5. Future research

For future research, it would be interesting to test the proposed attributes and guidelines in more industrial cases, develop clearer criteria for assessing cloneability, and study safety validation and certification in more depth. It would also be relevant to examine the economic

side more clearly, especially the balance between higher upfront engineering effort and later savings through repeated reuse. Also, It would be interesting to investigate how machine learning could support the development of cloneable automation. The main point of this thesis is that the most realistic path towards cloneable automation is a stable, documented, verified, and usable core that can be deployed repeatedly within defined boundaries.

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Appendix: Interview Guide

Intervjuguide:
Klonbara automationssystem, idé, semistrukturerad intervju med följdfrågor som kan användas för att fördjupa frågorna vid behov.
Obs: Led inte intervjun med klonbarhet

Idé för att presentera studien för intervjuobjektet:
I min masteruppsats studerar jag hur automationslösningar kan återanvändas mellan projekt, vad som återanvänds idag, vad som är svårt att återanvända och vad som skulle göra återanvändning snabbare och mer tillförlitlig.

Intervjuinformation:
Innan vi börjar, är det okej om jag spelar in intervjun? Inspelningen kommer endast att användas för denna studie och av mig, den kommer inte att delas med någon annan.
Jag skickar en skriftlig sammanfattning av vad jag har förstått från vårt samtal, så att du kan granska den innan jag använder den i arbetet.
Du kommer att vara anonym i rapporten

0. **Uppvärmning:**
   Vad har du för roll och vilka typer av automationsprojekt arbetar du vanligtvis med?

1. **Återanvändning i praktiken**
   Beskriv ett nyligen genomfört automationsprojekt. Återanvände ni redan utvecklade lösningar från tidigare projekt, och varför (eller varför inte)?

   a. Mjukvara (PLC- och/eller robotkod, FB, bibliotek)

b. Konfiguration (recept, I/O-mappar, rörelseparametrar)
c. Mekaniska moduler (gripdon, fixturer)
d. Dokumentation (mallar, checklistor, krav)
e. Testprocedurer (FAT/SAT, valideringsrutiner)

2. Vad är svårast eller dyrast att utveckla i ett automationsprojekt?

a. Vilka delar i ett projekt brukar ta mest ingenjörstid och/eller integrationstid?

b. Vad brukar skapa överraskningar under idrifttagning?

3. Gränser

Vid driftsätter av en liknande lösning i ett nytt projekt, vad måste ändras och vad kan förbli samma?

a. Gripdon och/eller verktyg

b. Säkerhet

c. Standarder (t.ex. kundstandarder för namngivning, dokumentation osv.)

d. Integration mellan informations- och operativ teknik (IT/OT)

4. Befintliga system (legacy) som begränsning

Hur påverkar befintlig utrustning, standarder och kompetens vad ni kan återanvända?

a. Leverantörsekosystem (en leverantör vs flera leverantörer?)

b. Versioner (PLC- och/eller robot-FW, bibliotek, verktyg osv.)

c. Installerad bas, t.ex. legacy

d. Kompetens (tillgängliga färdigheter, förtroende hos människor)

5. Validering och certifieringspåslag

När du ändrar eller re konfigurerar en befintlig automationscell, vilka valideringssteg är nödvändiga?

a. Vilka förändringar triggar om testning?

b. Hur kan återvalidering undvikas eller minimeras men ändå uppfylla krav på efterlevnad och säkerhet?

6. **Möjliggörare:**
   Vad, enligt din erfarenhet, möjliggör snabbare driftsättning och integration när man återanvänder liknande automationslösningar?

   a. Modularitet
   b. Parameterisering
   c. Plug and Produce (PnP)
   d. Datautbyte, till exempel Automation Markup Language (AML)

7. **Mänskliga aspekter, upplevd komplexitet:**
   Vad gör en lösning enkel respektive komplex för operatörer och underhåll?

   a. Användargränssnitt (UI) – mönster och/eller mallar (operatörernas vy? Tar ni in operatörernas önskemål/idéer?)
   b. Utbildning (självförklarande? ”Minska upplevd komplexitet”)
   c. Hantering av genvägar/workarounds, till exempel säkerhet. (Kan detta hanteras från er sida?)

8. **Önskefråga:**
   Om du fick ändra en sak för att förbättra återanvändning mellan projekt, vad skulle det vara?

   a. Vad skulle den förändringen möjliggöra?

9. **Klonbart automationssystem:**
   Vad tror du att ett koncept för klonbart automationssystem skulle kunna innebära?

   a. Hur skulle du definiera det?

   b. Har ni redan den här typen av lösning?

   c. Är er återanvändning idag mer ”kopiera & anpassa” eller ”instans & konfiguration”, och varför?